

PROJECT REPORT

EXPERIMENT - 11

Project Statement

Design and develop a Cloud-Based Industrial DG Monitoring and Analytics System. The system collects and analyzes diesel generator performance data using cloud technologies for remote monitoring applications.

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CLOUD-BASED INDUSTRIAL DG MONITORING AND ANALYTICS SYSTEM USING ESP8266

Project Overview

The Cloud-Based Industrial Diesel Generator (DG) Monitoring and Analytics System is an IoT-based monitoring system designed to collect important operating parameters of a diesel generator and make the data available for remote monitoring and analysis.

The system uses an ESP8266 Node MCU as the main controller. An ACS712 Current Sensor is used to monitor electrical current, while a DHT11 sensor measures temperature and humidity around the generator environment. The collected sensor data is processed by the Node MCU and transmitted through Wi-Fi to a cloud platform for remote monitoring.

This system helps industries monitor DG operating conditions without requiring continuous physical inspection. Cloud-based data storage also makes it possible to observe trends and identify abnormal operating conditions.

Main Objectives:

The main objectives of the project are:

- To develop an IoT-based DG monitoring system.
- To measure the current consumed by the DG load using the ACS712 sensor.
- To monitor temperature and humidity using the DHT11 sensor.
- To process sensor readings using ESP8266 Node MCU.
- To transmit DG monitoring data through Wi-Fi.
- To store sensor data on a cloud platform.
- To enable remote monitoring of DG operating conditions.
- To analyze collected data for identifying abnormal conditions.

- To reduce the need for continuous manual monitoring.
- To provide a foundation for predictive and preventive maintenance.

Problem Statement

Design and develop a Cloud-Based Industrial DG Monitoring and Analytics System. The system collects and analyzes diesel generator performance data using cloud technologies for remote monitoring applications.

Industrial diesel generators require regular monitoring to ensure reliable operation. Parameters such as electrical current and surrounding temperature can provide useful information about generator operating conditions. Conventional monitoring methods may require personnel to inspect the generator locally and record readings manually.

The proposed system solves this problem by using IoT sensors and an ESP8266 Node MCU to collect generator-related data and transmit it to a cloud platform. The information can then be viewed remotely and analyzed for monitoring and maintenance purposes.

Problems Addressed:

The proposed system addresses the following problems:

- Lack of continuous DG parameter monitoring.
- Dependence on manual inspection.
- Difficulty in monitoring generators located in remote areas.
- Delayed identification of abnormal operating conditions.
- Lack of centralized historical sensor data.
- Difficulty in analyzing generator operating trends.
- Increased maintenance effort.
- Limited accessibility to real-time operating information.

Proposed Solution

The proposed system uses an ESP8266 Node MCU as the central IoT controller.

The ACS712 Current Sensor measures the current associated with the monitored DG load. The DHT11 sensor measures temperature and humidity. The ESP8266 reads the sensor values, processes the data and connects to a Wi-Fi network.

The measured parameters are uploaded to a cloud monitoring platform. The cloud platform can display the data using numerical values, graphs and historical records. This allows authorized users to monitor DG conditions remotely.

Proposed Operation:

The proposed Cloud-Based Industrial DG Monitoring and Analytics System operates by continuously collecting important operating and environmental parameters of a diesel generator and transmitting the information to a cloud platform for remote monitoring.

- **System Initialization:** The ESP8266 Node MCU initializes the ACS712 current sensor and DHT11 temperature and humidity sensor.
- **Current Monitoring:** The ACS712 measures the electrical current associated with the monitored DG load and provides an analog signal to the ESP8266.
- **Environmental Monitoring:** The DHT11 measures the temperature and humidity surrounding the generator.
- **Data Processing:** The ESP8266 reads and processes the sensor values at predefined time intervals.
- **Wi-Fi Communication:** The Node MCU connects to an available Wi-Fi network using its built-in wireless capability.
- **Cloud Data Transmission:** The processed current, temperature, and humidity values are transmitted to a cloud monitoring platform.
- **Cloud Storage:** The cloud platform stores the received data for future reference and analysis.
- **Remote Monitoring:** Authorized users can access the cloud dashboard from an Internet-connected device to monitor DG conditions remotely.

- **Performance Analysis:** Historical sensor data can be represented using graphs and trends to identify changes in generator operating conditions.
- **Continuous Operation:** The system repeats the sensing, processing, transmission, and storage process continuously at the configured sampling interval.

System Architecture

The Cloud-Based Industrial DG Monitoring and Analytics System is designed using a layered IoT architecture. The system collects diesel generator operating data through sensors, processes the data using the ESP8266 Node MCU, transfers it through Wi-Fi to a cloud platform, and provides the information to remote users for monitoring and analysis.

Input Section:

The Input Section consists of sensors used to collect DG-related parameters.

- **ACS712 Current Sensor:** Measures the electrical current associated with the monitored DG load.
- **DHT11 Sensor:** Measures the surrounding temperature and humidity of the generator.
- The sensor outputs are provided to the ESP8266 Node MCU for processing.

Processing Section:

The ESP8266 Node MCU acts as the central processing unit. It receives the sensor signals, converts them into usable values and prepares the data for cloud transmission.

Communication Section:

The ESP8266 uses its built-in Wi-Fi capability to connect the monitoring system to the Internet.

Cloud Section:

The cloud platform receives the sensor readings and provides facilities for:

- Data storage
- Remote monitoring
- Graphical visualization
- Historical data analysis
- Performance observation

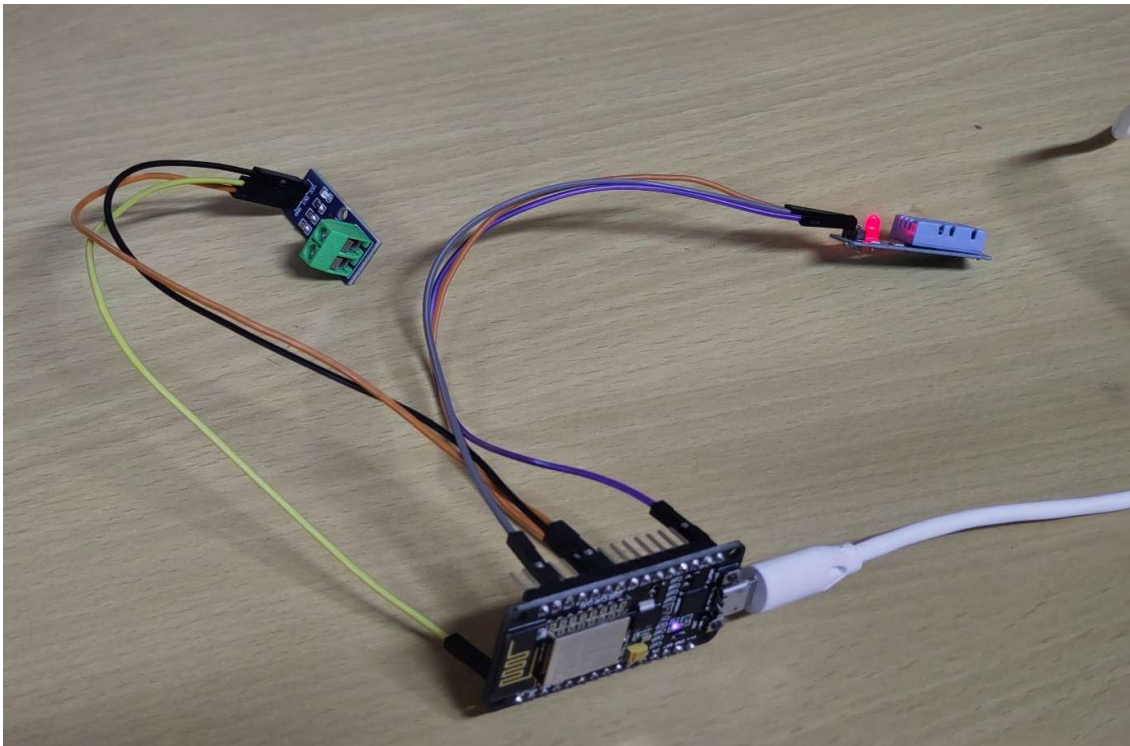
User Section:

The user can remotely access the cloud dashboard using a suitable Internet-connected device.

Block Diagram:

DG Parameters → Sensors → ESP8266 Node MCU → Wi-Fi/Internet → Cloud Platform → Data Storage & Analytics → Remote User

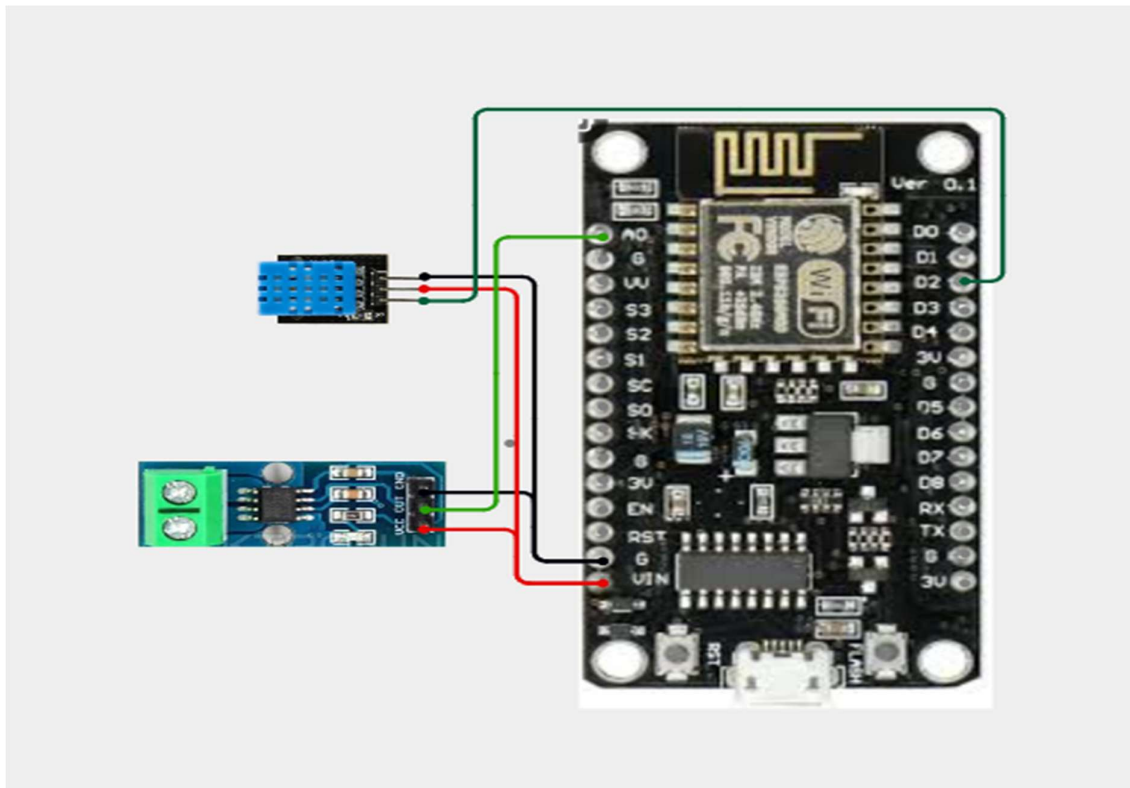
Image:



Circuit Table

S. No	Component	Pin	ESP8266 Node MCU Pin	Function
1	ACS712 Current Sensor	VCC	VIN / suitable supply	Provides power to the current sensor
2	ACS712	GND	GND	Common ground
3	ACS712	OUT	A0	Sends analog current-sensing signal
4	DHT11 Sensor	VCC	3.3V	Provides power to DHT11
5	DHT11	GND	GND	Common ground
6	DHT11	DATA	D4 (GPIO2)	Sends temperature and humidity data
7	ESP8266 Node MCU	USB	USB Power	Powers the Node MCU and prototype

Circuit Design



Important Pin Mapping:

The Cloud-Based Industrial DG Monitoring and Analytics System uses the following important pin connections:

- **ACS712 Current Sensor OUT → ESP8266 A0**
Used to read the analog current measurement from the ACS712 sensor.
- **ACS712 VCC → VIN / Suitable Supply**
Provides the required power to the ACS712 module according to its specifications.
- **ACS712 GND → ESP8266 GND**
Provides a common ground connection.
- **DHT11 DATA → ESP8266 D4 (GPIO2)**
Used to receive temperature and humidity data from the DHT11 sensor.
- **DHT11 VCC → ESP8266 3.3V**
Provides power to the DHT11 sensor.
- **DHT11 GND → ESP8266 GND**
Provides a common ground connection.
- **ESP8266 Wi-Fi → Internet/Cloud Platform**
The built-in Wi-Fi of the ESP8266 is used to transmit the collected DG monitoring data to the cloud.

Algorithm

- Start
- Initialize the ESP8266 Node MCU.
- Initialize the ACS712 Current Sensor.
- Initialize the DHT11 Temperature and Humidity Sensor.
- Connect the ESP8266 to the available Wi-Fi network.
- Check whether the Wi-Fi connection is successful.
- Read the current value from the ACS712 sensor through A0.
- Read the temperature and humidity values from the DHT11 sensor through D4.
- Process and calibrate the sensor readings.

- Check whether the sensor readings are valid.
- Prepare the collected data for transmission.
- Send the current, temperature, and humidity data to the cloud platform.
- Store the received data in the cloud.
- Display the data on the cloud dashboard.
- Analyze the stored data using graphs and trends.
- Check for abnormal changes in the monitored parameters.
- Wait for the specified monitoring interval.
- Repeat the process from Step 7.
- Stop when the system is switched OFF.

Coding

```
#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

#include <DHT.h>

// Connected to GPIO4 (D2 on NodeMCU/Wemos D1 Mini)

#define DHTPIN 4

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

// WiFi Credentials

const char* ssid = "Rithick";

const char* password = "12345678";

// ThingSpeak Configuration

unsigned long channelID = 3462452;
```

```
const char *writeAPIKey = "TS4L9164ENORL68P";
```

```
WiFiClient client;
```

```
// ACS712 Current Sensor connected to Analog Pin A0
```

```
const int sensorPin = A0;
```

```
void setup()
```

```
{
```

```
  Serial.begin(115200);
```

```
  dht.begin();
```

```
  delay(2000); // Allow DHT11 sensor time to stabilize
```

```
  WiFi.begin(ssid, password);
```

```
  Serial.print("Connecting to WiFi");
```

```
  while (WiFi.status() != WL_CONNECTED)
```

```
  {
```

```
    delay(500);
```

```
    Serial.print(".");
```

```
  }
```

```
  Serial.println();
```

```
  Serial.println("WiFi Connected");
```

```
ThingSpeak.begin(client);
}

void loop()
{
    // Read DHT11 sensor data
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    // Retry once if reading fails
    if (isnan(temperature) || isnan(humidity))
    {
        delay(1000);
        temperature = dht.readTemperature();
        humidity = dht.readHumidity();
    }

    if (isnan(temperature) || isnan(humidity))
    {
        Serial.println("DHT Read Failed!");
        delay(5000);
        return;
    }

    // Read raw ADC value from ACS712
```

```
int adc = analogRead(sensorPin);  
float current = adc;  
  
Serial.println("-----");  
Serial.print("Temperature : ");  
Serial.print(temperature);  
Serial.println(" C");  
  
Serial.print("Humidity  : ");  
Serial.print(humidity);  
Serial.println(" %");  
  
Serial.print("ACS712 Raw : ");  
Serial.println(adc);  
  
// Prepare ThingSpeak fields  
ThingSpeak.setField(1, current);  
ThingSpeak.setField(2, temperature);  
ThingSpeak.setField(3, humidity);  
  
// Send data to ThingSpeak  
int status = ThingSpeak.writeFields(channelID, writeAPIKey);  
if (status == 200)  
{  
    Serial.println("ThingSpeak Upload Success");  
}
```

```

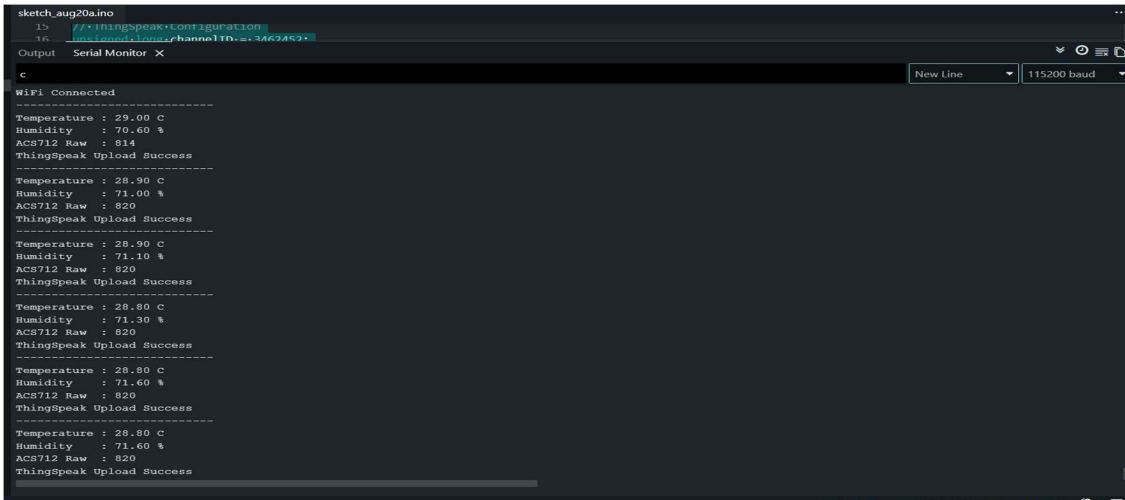
else
{
    Serial.print("ThingSpeak Error: ");

    Serial.println(status);
}

// Wait 20 seconds before the next reading (ThingSpeak requires >= 15 seconds)
delay(20000);
}

```

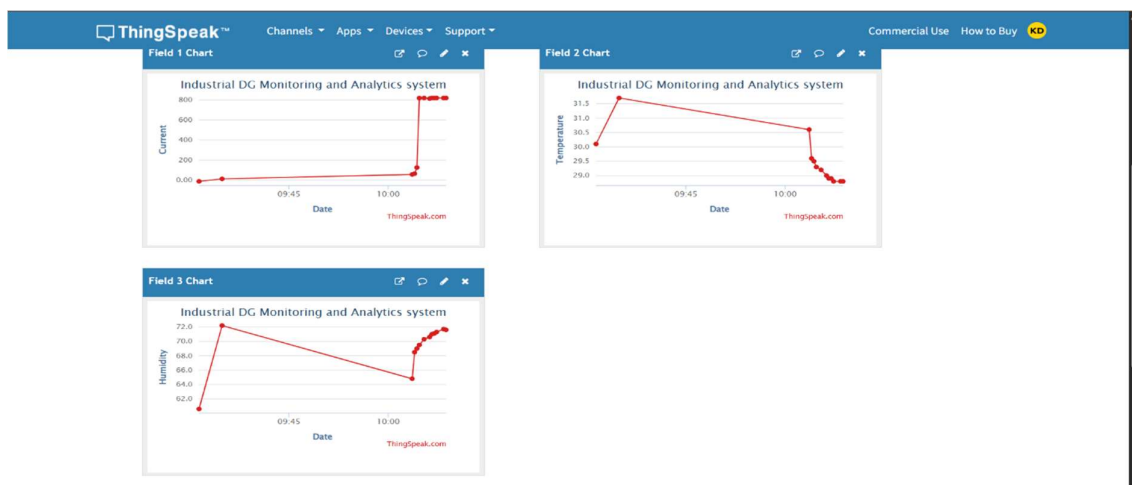
Coding Output:



```

sketch_aug20a.ino
15: //ThingSpeak Configuration
16: #include <ThingSpeak.h>
Output Serial Monitor x
c
WiFi Connected
-----
Temperature : 29.00 C
Humidity : 70.60 %
ACS712 Raw : 814
ThingSpeak Upload Success
-----
Temperature : 28.90 C
Humidity : 71.00 %
ACS712 Raw : 820
ThingSpeak Upload Success
-----
Temperature : 28.90 C
Humidity : 71.10 %
ACS712 Raw : 820
ThingSpeak Upload Success
-----
Temperature : 28.80 C
Humidity : 71.30 %
ACS712 Raw : 820
ThingSpeak Upload Success
-----
Temperature : 28.80 C
Humidity : 71.60 %
ACS712 Raw : 820
ThingSpeak Upload Success
-----
Temperature : 28.80 C
Humidity : 71.60 %
ACS712 Raw : 820
ThingSpeak Upload Success
-----
Ln 104, Col 2 - Generic ESP8266 Module on COM10

```



System Working

The Cloud-Based Industrial DG Monitoring and Analytics System works by continuously collecting and transmitting important DG operating and environmental parameters to a cloud platform for remote monitoring.

- **System Power ON:** When the system is powered ON, the ESP8266 Node MCU initializes the ACS712 and DHT11 sensors.
- **Current Measurement:** The ACS712 Current Sensor measures the electrical current of the monitored DG load. Its analog output is provided to the **A0 pin** of the ESP8266.
- **Temperature and Humidity Measurement:** The DHT11 sensor measures the temperature and humidity surrounding the generator and sends the data to the **D4 (GPIO2)** pin of the ESP8266.
- **Data Processing:** The ESP8266 reads the sensor values, processes them, and prepares the measurements for transmission.
- **Wi-Fi Connection:** The ESP8266 uses its built-in Wi-Fi capability to connect to an available Internet network.
- **Cloud Data Transmission:** The processed current, temperature, and humidity values are transmitted to the selected cloud platform at regular intervals.
- **Cloud Storage:** The cloud platform stores the received measurements along with their corresponding time information.
- **Remote Monitoring:** The stored data is displayed on a cloud dashboard. Users can remotely observe current, temperature, and humidity values using an Internet-connected device.
- **Data Analysis:** Historical readings can be displayed as graphs and trends. Changes in the monitored parameters can help identify unusual operating conditions and support maintenance decisions.
- **Continuous Monitoring:** After completing one measurement and transmission cycle, the system waits for the configured interval and repeats the process continuously.

Bill of Materials

S. No	Component	Specification / Description	Quantity
1	ESP8266 Node MCU	Wi-Fi-enabled microcontroller development board	1
2	ACS712 Current Sensor	Hall-effect current sensing module	1
3	DHT11 Sensor	Temperature and humidity sensor	1
4	Breadboard	Solderless prototyping board	1
5	Jumper Wires	Male-to-male / suitable connecting wires	1 Set

Prototype Implementation

The prototype of the Cloud-Based Industrial DG Monitoring and Analytics System is implemented using an ESP8266 Node MCU, ACS712 Current Sensor, DHT11 Sensor, breadboard, and jumper wires. The prototype demonstrates the collection, processing, and cloud-based monitoring of DG-related parameters.

Hardware Assembly:

The ESP8266 Node MCU is placed on the breadboard and connected to the sensors using jumper wires.

- The ACS712 OUT pin is connected to A0 of the ESP8266.
- The DHT11 DATA pin is connected to D4 (GPIO2).
- The sensor grounds are connected to the GND of the Node MCU.
- The DHT11 is supplied from the appropriate 3.3 V connection.
- The ACS712 is powered according to the specification of the particular module being used.

Sensor Installation:

The ACS712 is positioned in the current-sensing path of a safe, isolated low-voltage test load for prototype demonstration. The DHT11 is positioned near the generator environment to measure temperature and humidity.

ESP8266 Programming:

The ESP8266 is programmed using the Arduino IDE. The program is configured to:

- Initialize the ACS712 and DHT11.
- Read current, temperature, and humidity.
- Connect to the Wi-Fi network.
- Process the sensor readings.
- Send the collected data to the cloud platform at regular intervals.

Cloud Integration:

A suitable IoT cloud platform such as Thing Speak can be configured to receive the sensor data.

Separate data fields can be assigned for:

- **Field 1:** Current
- **Field 2:** Temperature
- **Field 3:** Humidity

The cloud platform stores the received values and generates graphs for monitoring parameter variations over time.

Prototype Testing:

After completing the hardware and software setup:

- Power the ESP8266.
- Establish the Wi-Fi connection.
- Start the monitoring program.
- Verify current readings from the ACS712.

- Verify temperature and humidity readings from the DHT11.
- Check whether data is successfully uploaded to the cloud.
- Observe the cloud dashboard.
- Change the safe test-load condition and observe the corresponding current variation.
- Observe changes in temperature and humidity over time.

Prototype Operation:

The complete prototype operates as:

Sensors → ESP8266 → Wi-Fi → Cloud Platform → Dashboard → Remote Monitoring

Prototype Working

The Cloud-Based Industrial DG Monitoring and Analytics System prototype works by continuously collecting electrical and environmental parameters using the ACS712 Current Sensor and DHT11 Sensor. The ACS712 measures the current of a suitable monitored load and provides an analog signal to the A0 pin of the ESP8266 Node MCU, while the DHT11 measures the surrounding temperature and humidity and sends the data through D4 (GPIO2). The ESP8266 processes the sensor readings and connects to the Internet through its built-in Wi-Fi capability. The processed current, temperature, and humidity values are periodically transmitted to a cloud platform, where the data is stored and displayed through a dashboard using numerical values and graphs. Users can remotely monitor the collected parameters and analyze historical trends to identify changes in DG operating conditions. The system continuously repeats the sensing, processing, cloud transmission, and monitoring process, providing a simple IoT-based solution for remote DG performance monitoring. For prototype testing, the ACS712 should be connected to a suitable safe, isolated test load rather than directly to high-voltage generator output.

Expected Prototype Output

S. No	Parameter	Sensor/Source	Expected Output
1	DG Current	ACS712 Current Sensor	Current value in A
2	Temperature	DHT11	Temperature in °C
3	Humidity	DHT11	Relative humidity in %
4	Wi-Fi Status	ESP8266	Connected / Disconnected
5	Cloud Status	ESP8266 + Cloud	Data Uploaded / Failed

Prototype Demonstration

The prototype demonstration of the Cloud-Based Industrial DG Monitoring and Analytics System begins by powering the ESP8266 Node MCU and initializing the ACS712 Current Sensor and DHT11 Temperature and Humidity Sensor. The ACS712 is connected to a suitable safe test load to demonstrate current measurement, while the DHT11 monitors the surrounding temperature and humidity. The ESP8266 reads the sensor values and displays them through the Serial Monitor during initial testing. After establishing a Wi-Fi connection, the Node MCU transmits the collected current, temperature, and humidity data to the configured cloud platform. The cloud dashboard displays the received values and generates graphs showing parameter variations over time. During the demonstration, the test-load condition can be safely varied to observe changes in the current reading, while changes in the surrounding environment can be observed through the DHT11 readings. The demonstration confirms the complete data flow from sensor → ESP8266 → Wi-Fi → cloud → dashboard, showing the capability of the prototype for remote DG monitoring and basic performance analysis.

Results & Performance Analysis

The developed **Cloud-Based Industrial DG Monitoring and Analytics System** successfully demonstrates the collection and remote monitoring of important DG-related parameters using IoT technology. The ESP8266 Node MCU receives current data from the ACS712 Current Sensor and temperature and humidity data from the DHT11 Sensor. The collected values are processed by the Node MCU and transmitted through Wi-Fi to the cloud platform. The cloud dashboard provides remote visualization of the sensor readings and allows historical data to be observed through graphs. The prototype demonstrates that DG-related operating information can be monitored without continuous manual inspection.

Performance Analysis

Parameter	Performance
Current Monitoring	Continuous monitoring using ACS712
Temperature Monitoring	Periodic monitoring using DHT11
Humidity Monitoring	Periodic monitoring using DHT11
Data Processing	Performed by ESP8266 Node MCU
Wireless Communication	Wi-Fi based
Cloud Monitoring	Remote data visualization
Data Storage	Historical cloud-based storage
Trend Analysis	Graph-based monitoring
System Scalability	Additional sensors can be integrated